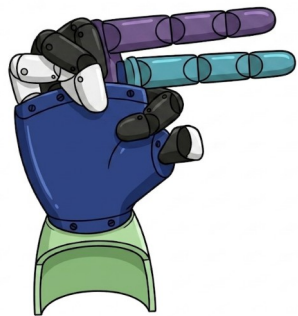




BuildBuddy @ Dartmouth

***II-Powered ASL Fingerspelling Hand
Building Instructions***
Designed for Dartmouth Students!





H



E



L



L

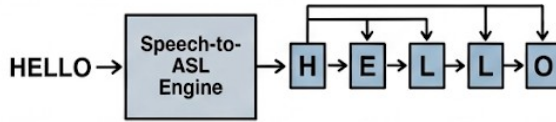


O

1. INPUT:



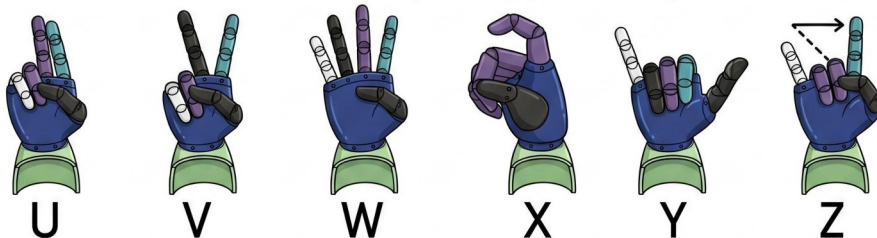
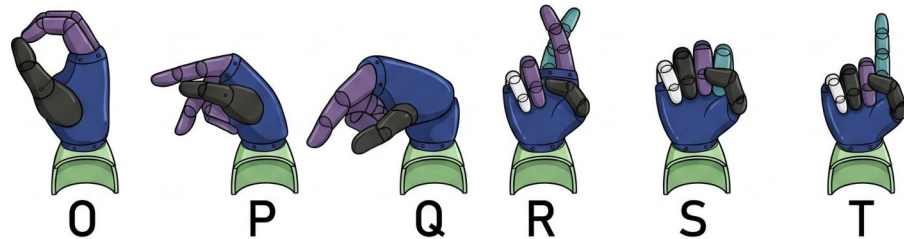
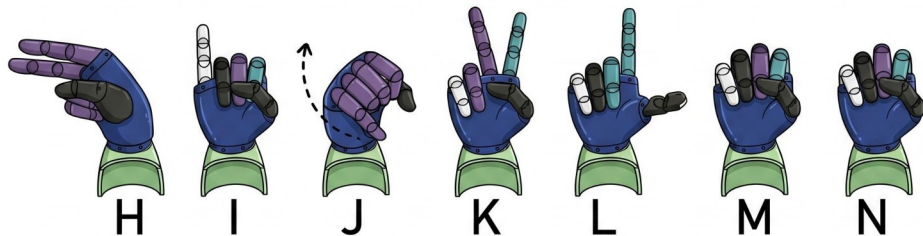
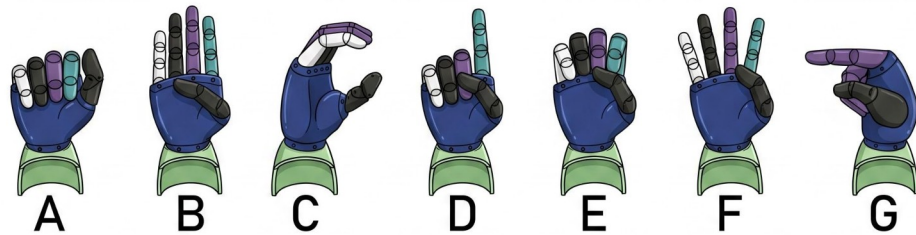
2. INTERPRETATION:



3. OUTPUT (SIGNING):



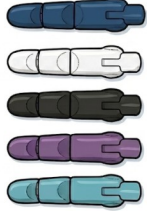
Build a 3D-printed
hand that turns
voice input into the
ASL alphabet
in 24 hours!



**Learn the ASL
alphabet!
Learn to build!**



Parts Needed



3D-printed hand parts
(see next page)



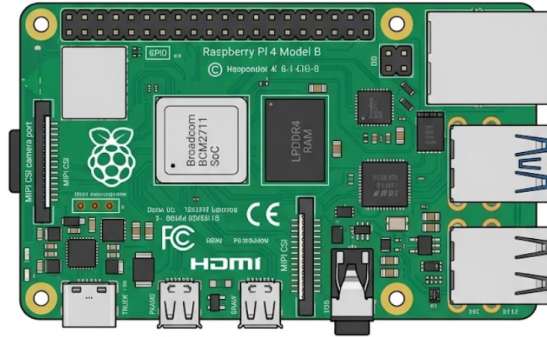
Bungee Cord



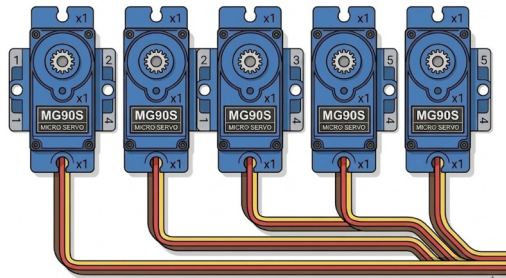
Fishing Line



Male-to-Female
Dupont Jumper Wires



Raspberry Pi 4 Model B



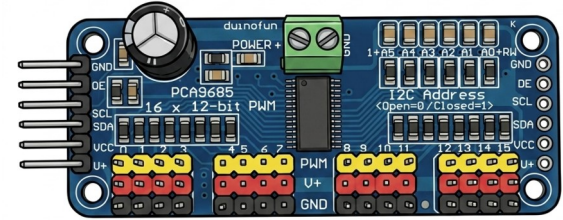
5x MG90S Micro Servos



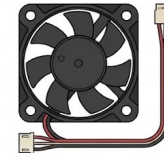
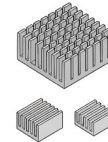
5V 5A Power Supply
with Barrel Jack Adapter



USB
Microphone



PCA9685



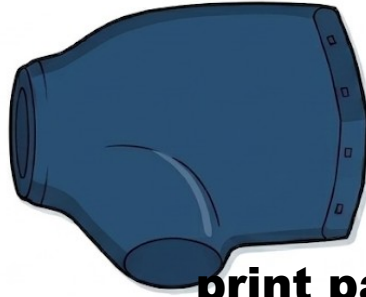
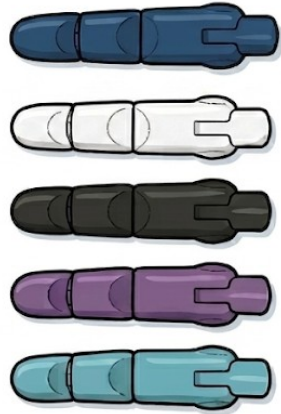
Heatsinks and Fan
for Pi



USB-C Power
Supply for Pi



3D-Printed Hand Parts



print parts from <https://tinyurl.com/2v6u7m3e>

Download and



3D-Printing @ Dartmouth

Go to MShop (C027) or Harold Edward Cable MakerSpace (ECSC 003), give the STL files from the previous page, and get them printed!



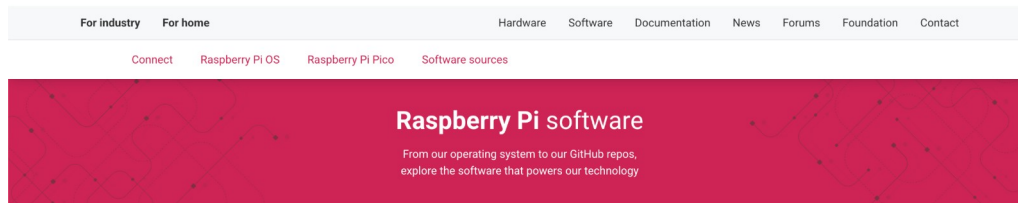


Part

Set Up the Raspberry Pi

Download Raspberry Pi Imager from [raspberrypi.com/software](https://www.raspberrypi.com/software)

1



Raspberry Pi Imager

Raspberry Pi Imager is the quick and easy way to install **Raspberry Pi OS** and other operating systems to a microSD card, ready to use with your Raspberry Pi.

Download and install Raspberry Pi Imager on a computer with an SD card reader. Insert the microSD card you'll use with your Raspberry Pi into the reader and run Raspberry Pi Imager.

[Download for macOS](#)

[Read the documentation →](#)

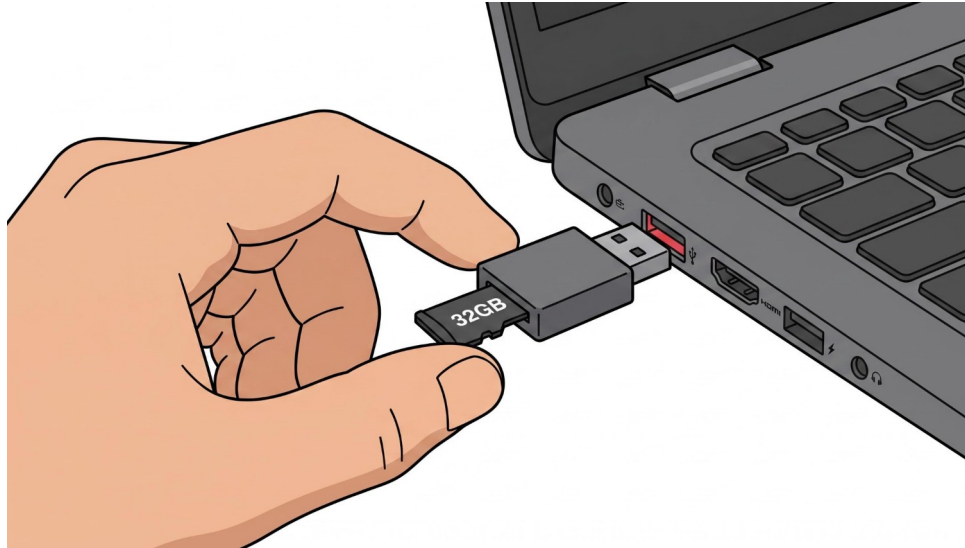
[Download for Windows](#)

[Download for Linux \(x86_64\)](#)



Plug in your 32 GB micro SD card into your laptop
(you may need a micro SD to SD card adapter)

2



Open Raspberry Pi Imager

Choose Device -> Raspberry Pi 4

3

Setup steps

Device

OS

Storage

Customisation

Writing

Done

APP OPTIONS

Select your Raspberry Pi device



Raspberry Pi 5

Raspberry Pi 5, 500 / 500+, and Compute Module 5



Raspberry Pi 4

Raspberry Pi 4 Model B, 400, and Compute Module 4 / 4S



Raspberry Pi 3

Raspberry Pi 3 Model A+ / B / B+ and Compute Module 3 / 3+



Raspberry Pi 2

Raspberry Pi 2 Model B



Raspberry Pi 1

Raspberry Pi 1 Model A / A+ / B / B+ and Compute Module 1

NEXT



Open Raspberry Pi Imager

Choose OS -> Raspberry Pi OS
(64-bit)

4

Setup steps

Device

OS

Storage

Customisation

Writing

Done

APP OPTIONS

Choose operating system

Select an operating system to install on your Raspberry Pi



Raspberry Pi OS (64-bit)

A port of Debian Trixie with the Raspberry Pi Desktop
(Recommended)

Cached on your computer

Released: 2026-04-21



Raspberry Pi OS (32-bit)

A port of Debian Trixie with the Raspberry Pi Desktop

Online - 1.2 GB download

Released: 2026-04-21



Raspberry Pi OS (Legacy, 32-bit)

A port of Debian Bookworm with security updates and desktop
environment

Online - 1.2 GB download

Released: 2026-04-13



Raspberry Pi OS (other)

Other Raspberry Pi OS based images

BACK

NEXT



Open Raspberry Pi Imager

Choose Storage -> {your micro SD card}

5

Setup steps

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

APP OPTIONS

Select your storage device



Apple SDXC Reader Media

28.2 GB

Mounted as /Volumes/bootfs



Exclude system drives

BACK

NEXT



Open Raspberry Pi Imager

Customisation → hostname →
“aslhand”

6

Device

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

APP OPTIONS

Customisation: Choose hostname

Enter your hostname

aslhand

A hostname is a unique name that identifies your Raspberry Pi on the network. It should contain only letters, numbers, and hyphens.

SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Customisation → Localisation

7

OS

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

APP OPTIONS

Customisation: Localisation

Select your location for suggested time zone and keyboard layout

Capital city:

Washington, D.C. (United States) ▼



Time zone:

America/New_York ▼

Keyboard layout:

us ▼

SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Customisation → User → Choose username

8

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Customisation: Choose username

Create a user account for your Raspberry Pi

Username:

Enter your username

ashleypi

Password:

Saved (hidden) – leave blank to keep

Confirm password:

Re-enter to change password

The username must be lowercase and contain only letters, numbers, underscores, and hyphens.

SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Customisation → Wi-Fi → Enter your Wi-Fi

9

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Customisation: Choose Wi-Fi

SECURE NETWORK

OPEN NETWORK

SSID:

Network name

KCC7

Password:

Saved (hidden) — leave blank to keep

Confirm password:

Re-enter to change password

☐ Hidden SSID

SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Customisation → SSH authentication

→ Use password authentication

10

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Customisation: SSH authentication

Configure SSH access

Enable SSH

[Learn about SSH](#)



Authentication mechanism:



Use password authentication



Use public key authentication

SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Customisation → Raspberry Pi
Connect → Off

11

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Customisation: Raspberry Pi Connect

Sign in to receive a token and enable Raspberry Pi Connect

Enable Raspberry Pi Connect

What is Raspberry Pi Connect?



SKIP CUSTOMISATION

BACK

NEXT



Open Raspberry Pi Imager

Writing → Write

12

Storage

Customisation

Hostname

Localisation

User

Wi-Fi

Remote access

Raspberry Pi Connect

Writing

Done

APP OPTIONS

Write image

Review your choices and write the image to the storage device

Summary

Device: **Raspberry Pi 4**
Operating system: **Raspberry Pi OS (64-bit)**
Storage: **Apple SDXC Reader Media**

Customisations to apply:

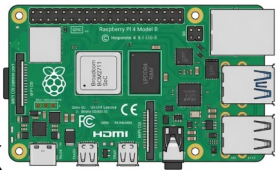
- Hostname configured
- Localisation configured
- User account configured
- Wi-Fi configured
- SSH enabled

BACK

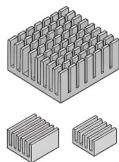
WRITE



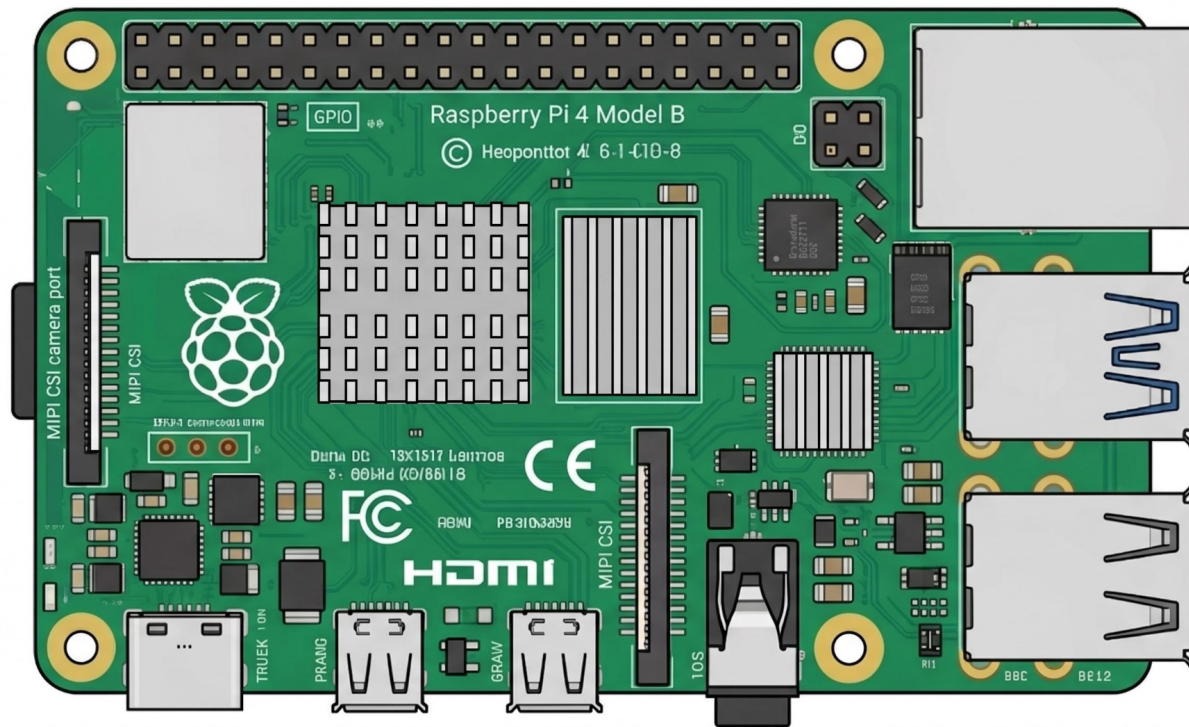
1x



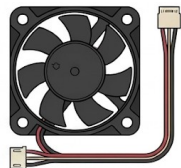
1x



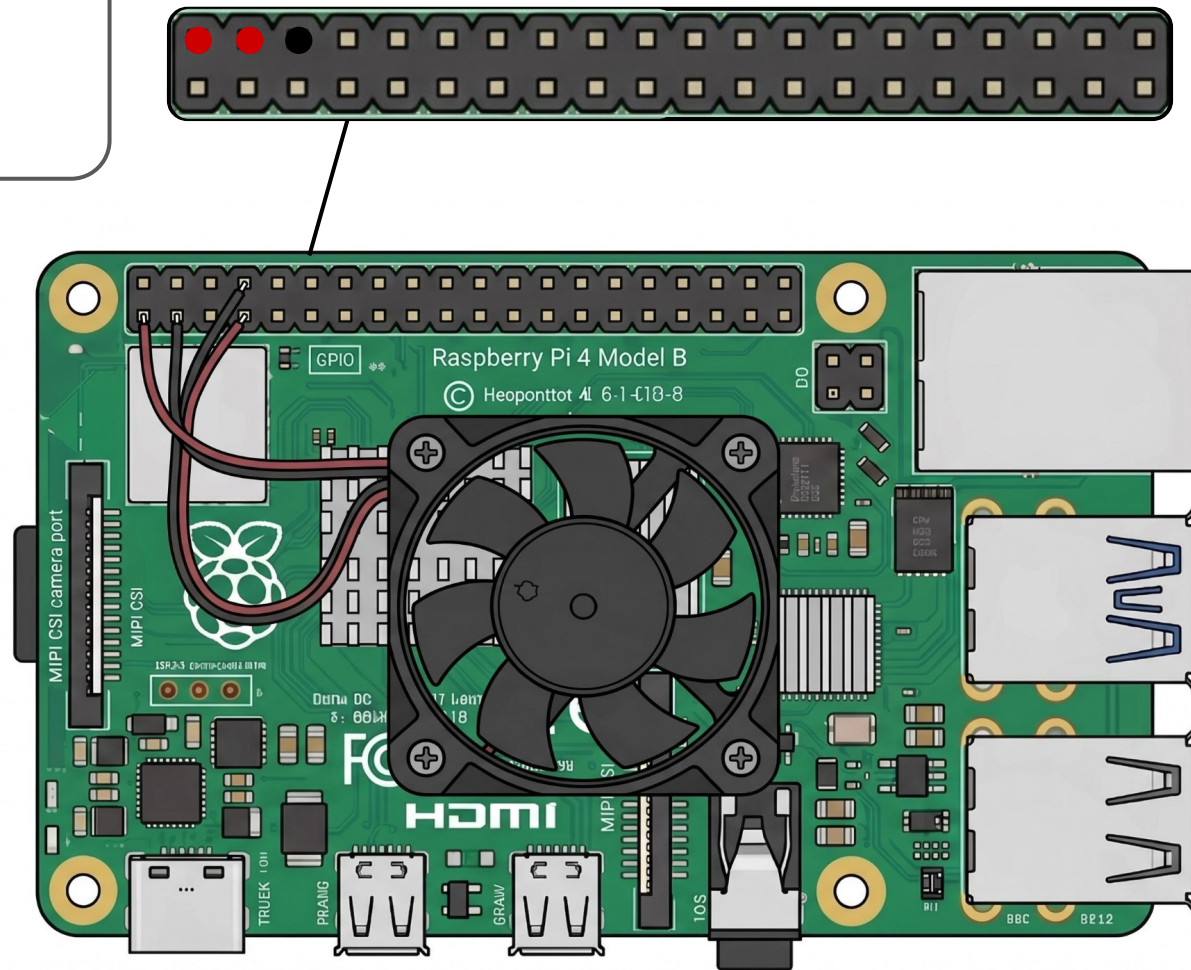
13



1x



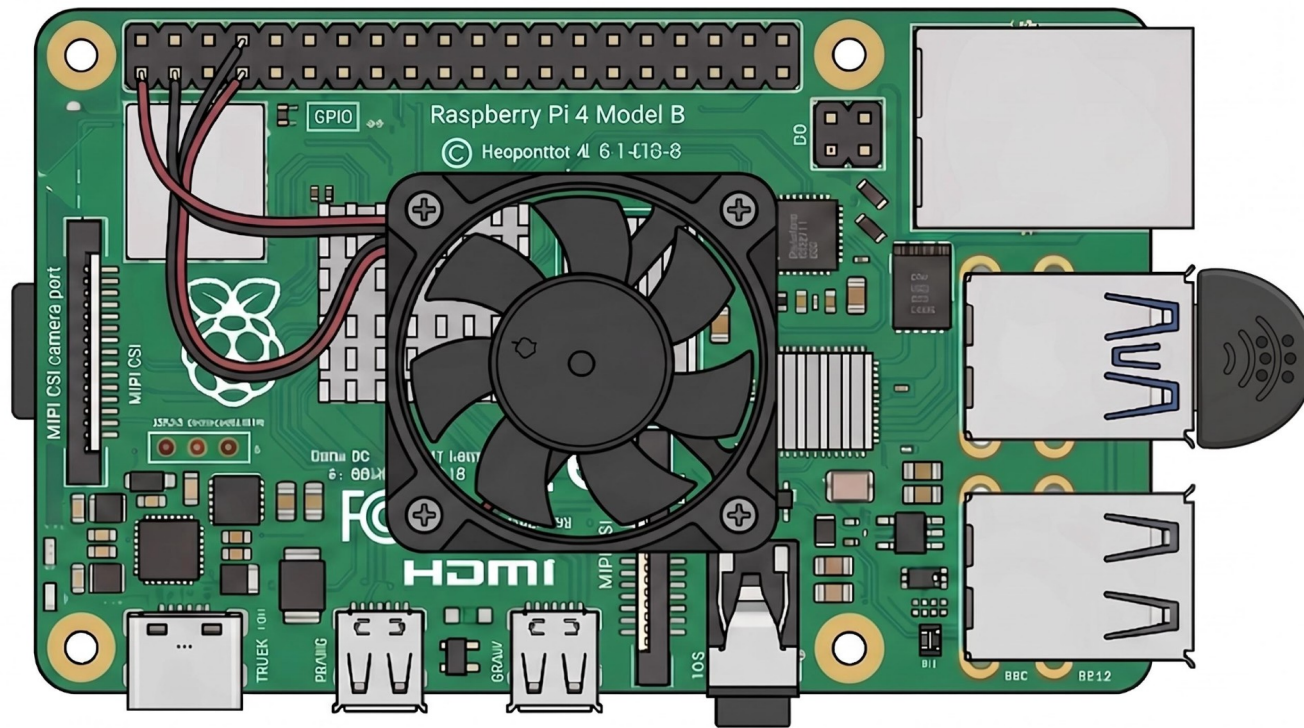
14



1x



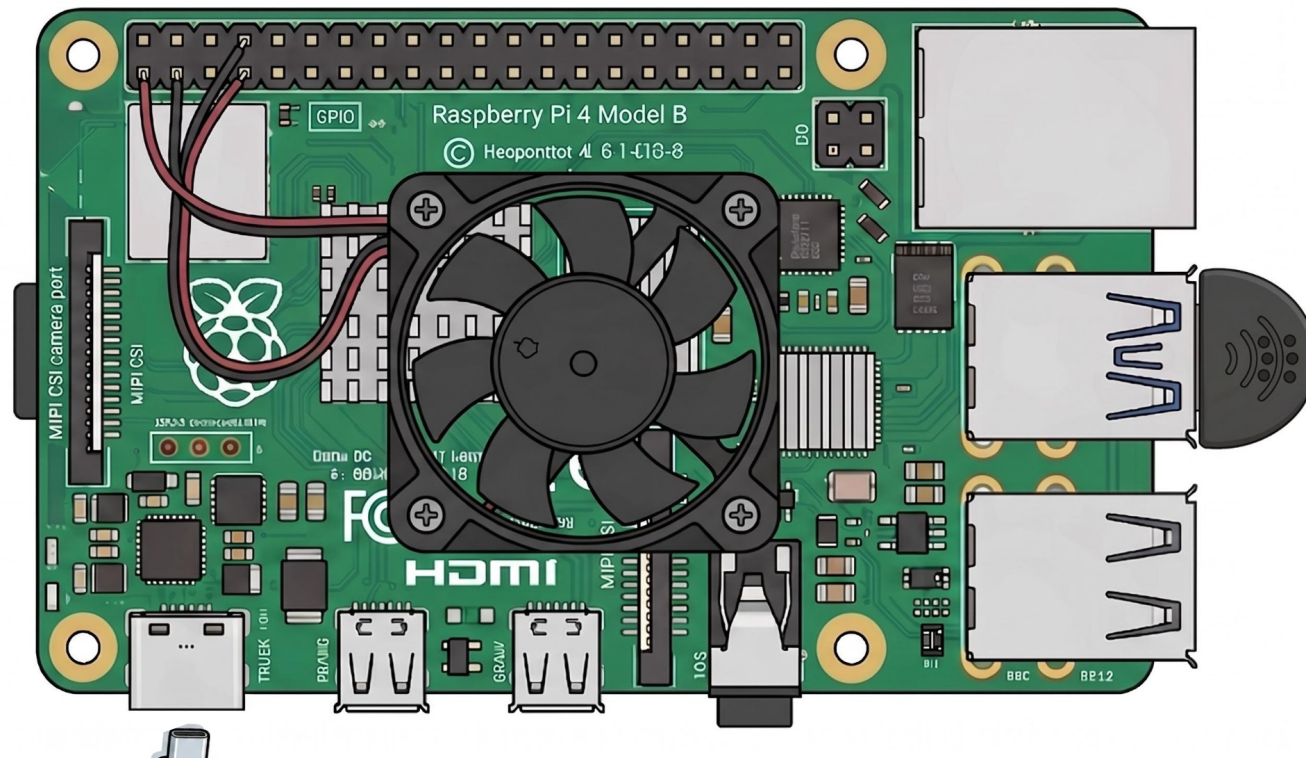
15



1x



16



Open Terminal on Mac

Connect to your Pi → ssh
{PiName}@aslhand.local

17

```
(base) yourName@yourLaptopName ~ %  
ssh {PiName}@aslhand.local
```



Open Terminal on Mac

Run these commands

18

```
ashleypi@aslhand.local:~ $ sudo apt update
ashleypi@aslhand.local:~ $ sudo apt upgrade
-y
ashleypi@aslhand.local:~ $ sudo apt install
```



Open Terminal on Mac

Run this command then “Interface Options” → “I2C” → Enable

19

```
ashleypi@aslhand.local:~ $ sudo raspi-  
confi
```



Open Terminal on Mac

Run this command

20

```
ashleypi@aslhand.local:~ $ pip3 install --break-  
system-packages adafruit-circuitpython-pca9685  
adafruit-circuitpython-servokit SpeechRecognition  
pyaudio
```



Open Terminal on Mac

Test the microphone

The last line should output ~860K if the mic is working

21

```
ashleypi@aslhand.local:~ $ arecord -l
ashleypi@aslhand.local:~ $ arecord -D plughw:1,0
-d 5 test.wav
ashleypi@aslhand.local:~ $ ls -lh test.wav
```



Par



Set Up the Hand

5x



1x



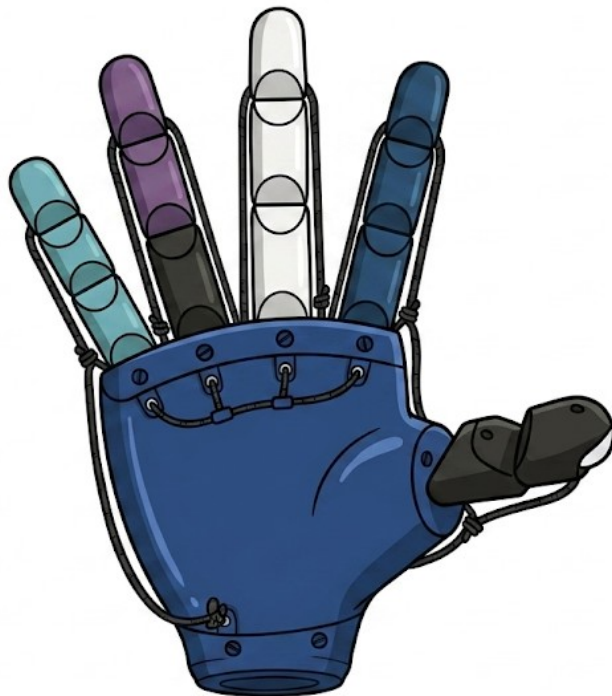
1



1x



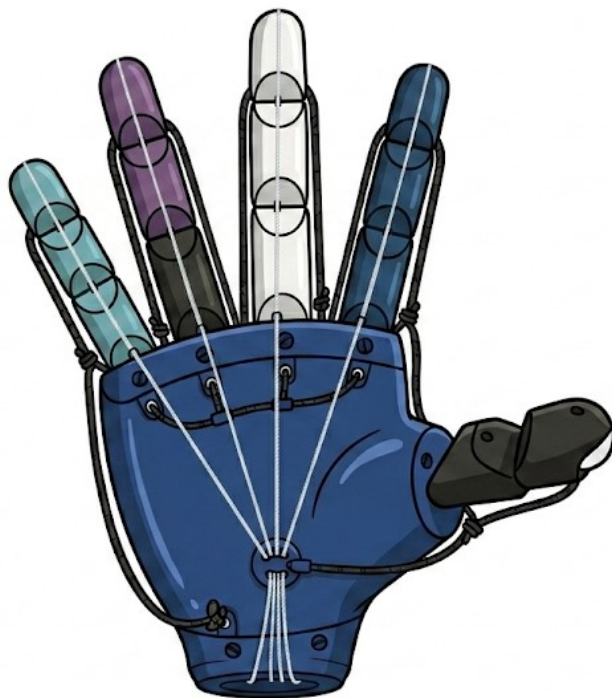
2



1x



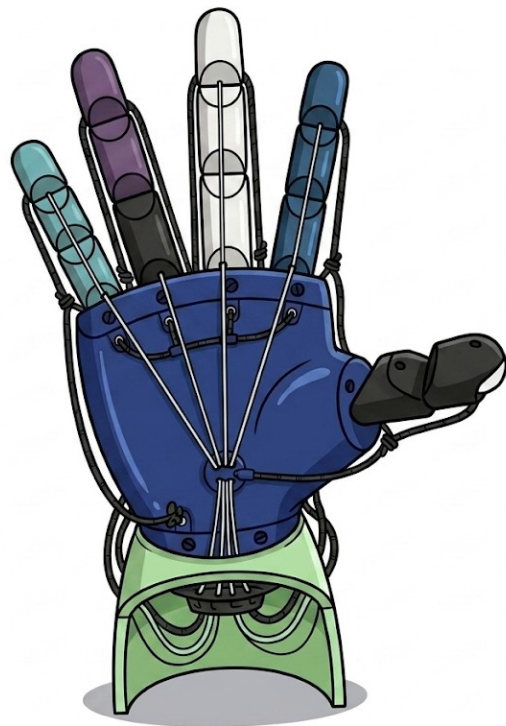
3



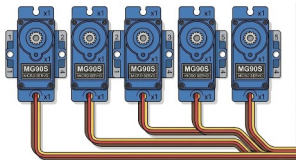
1x



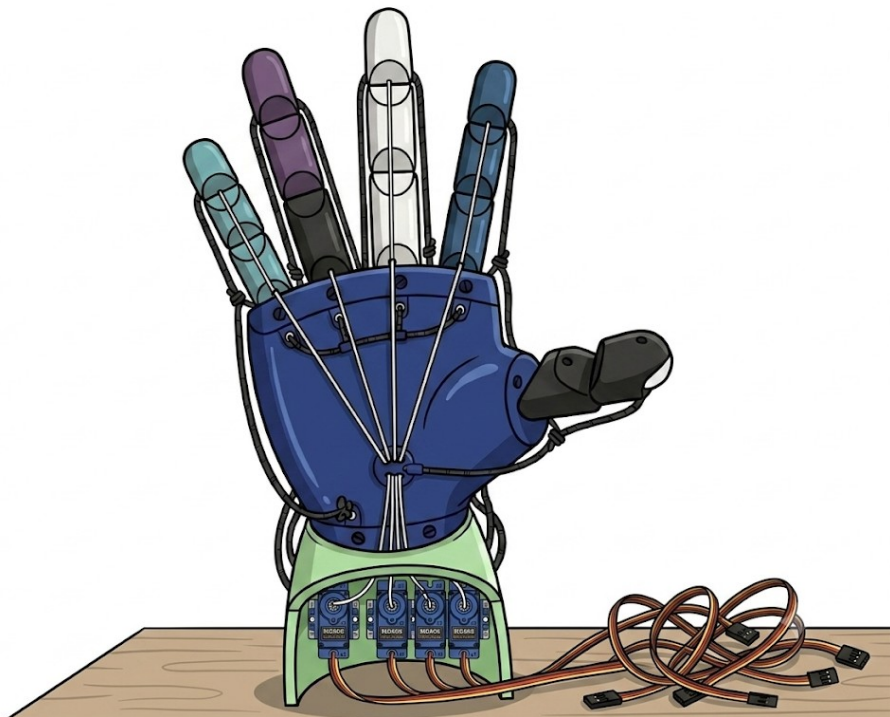
4



5x



5





Part

Connecting the Pi to the Hand

1x



4x



1

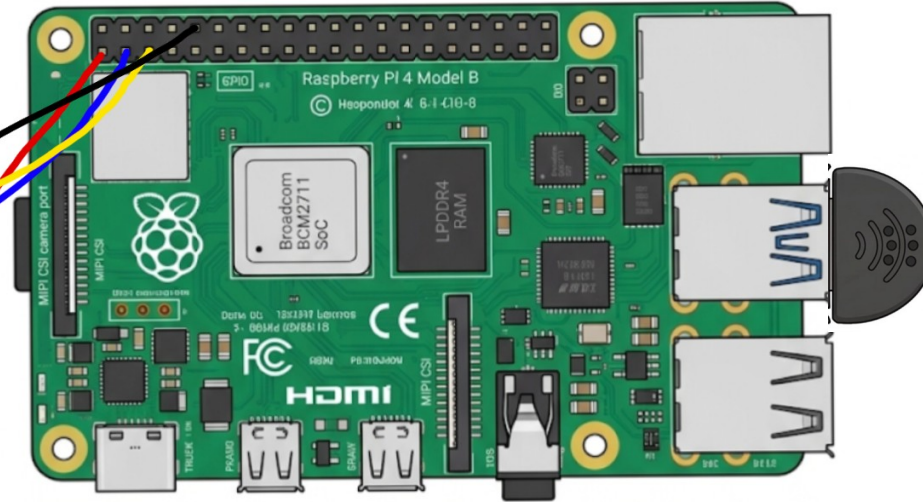
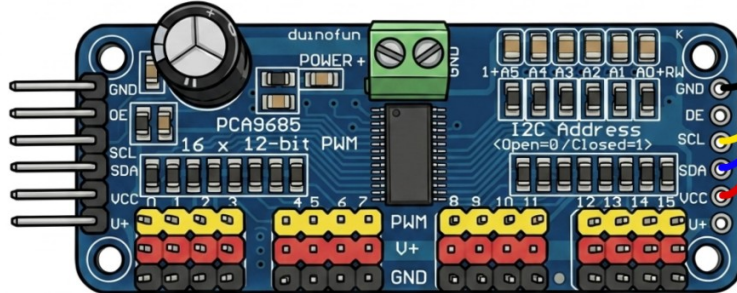
Raspberry Pi → PCA9685

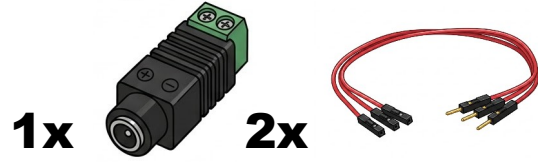
Pin 1 (3.3V) → VCC

Pin 9 (GND) → GND

Pin 3 (SDA GPIO 2) → SDA

Pin 5 (SCL GPIO 3) → SCL

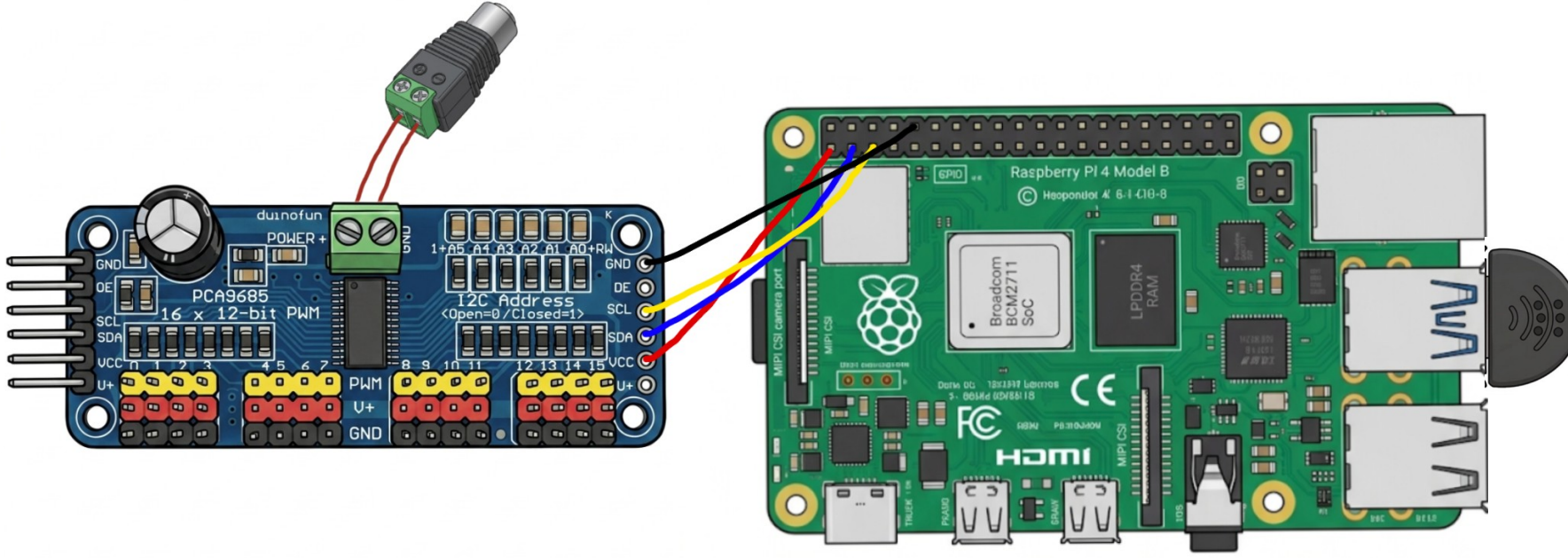




Barrel Jack Adapter → PCA9685

+ → V+
- → GND

2



A black, rectangular power supply unit (PSU) with a power cord and a connector cable. The power cord has a standard three-prong AC plug. The connector cable has a circular connector with a green tab, which is used to connect the PSU to the system unit.



4

MG90S → PCA9685

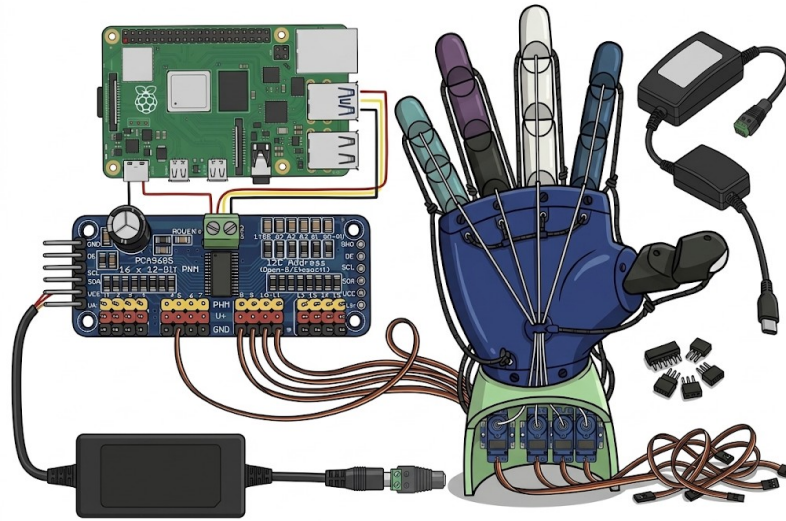
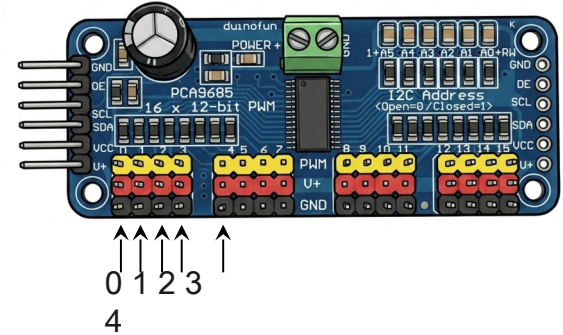
Thumb → Port 0

Index → Port 1

Middle → Port 2

Ring → Port 3

Pinky → Port 4



Make sure the yellow wire goes in the top row and the brown wire goes in the bottom row

Open Terminal on Mac

Run these commands

5

```
ashleypi@aslhand.local:~ $ nano  
asl_hand_nv
```



Open Terminal on Mac

Paste this script then Ctrl+O to save, Enter, then Ctrl+X to exit

6

```
https://tinyurl.com/  
5awuwn2v
```



Open Terminal on Mac

Run these commands

7

```
ashleypi@aslhand.local:~ $ python3  
asl_hand.py
```





YOU'RE DONE!